LAB INSEM OUTPUTS

QUESTION 1 OUTPUT

```
pragnya@DESKTOP-AUGV0MN:~$ nano banker.c
pragnya@DESKTOP-AUGV0MN:~$ gcc banker.c -o banker
pragnya@DESKTOP-AUGV0MN:~$ ./banker
Enter number of processes (max 10): 5
3
0 1 Enter number of resource types (max 10):
Enter Allocation Matrix (5 x 3):
0
2 0 0
3 0 2
2 1 1
0 0 2
7 5 3

Enter Maximum Claim Matrix (5 x 3):
3 2 2
9 0 2
2 2 2
4 3 3

Enter Available Resource Vector (3 values):
3 3 2

Need Matrix (Maximum - Allocation):
7 4 3
1 2 2
6 0 0
0 1 1
4 3 1
```

```
All Possible Safe Sequences:
Safe Sequence 1: P1 P3 P0 P2 P4
Safe Sequence 2: P1 P3 P0 P4 P2
Safe Sequence 3: P1 P3 P2 P0 P4
Safe Sequence 4: P1 P3 P2 P4 P0
Safe Sequence 5: P1 P3 P4 P0 P2
Safe Sequence 6: P1 P3 P4 P2 P0
Safe Sequence 7: P1 P4 P3 P0 P2
Safe Sequence 8: P1 P4 P3 P2 P0
Safe Sequence 9: P3 P1 P0 P2 P4
Safe Sequence 10: P3 P1 P0 P4 P2
Safe Sequence 11: P3 P1 P2 P0 P4
Safe Sequence 12: P3 P1 P2 P4 P0
Safe Sequence 13: P3 P1 P4 P0 P2
Safe Sequence 14: P3 P1 P4 P2 P0
Safe Sequence 15: P3 P4 P1 P0 P2
Safe Sequence 16: P3 P4 P1 P2 P0

System is in SAFE state.
Total safe sequences: 16
pragnya@DESKTOP-AUGV0MN:~$ ./banker
Enter number of processes (max 10): 3
1
1
2
Enter number of resource types (max 10):
Enter Allocation Matrix (3 x 1):
3
3
3
4
```

```
Enter Maximum Claim Matrix (3 x 1):

Enter Available Resource Vector (1 values):
0

Need Matrix (Maximum - Allocation):
2
1
1

All Possible Safe Sequences:

System is in UNSAFE state.
No complete safe sequence exists.
Processes unable to proceed initially: P0 P1 P2
Note: An unsafe state does not necessarily mean deadlock has already occurred.
pragnya@DESKTOP-AUGV0MN:~$
```

QUESTION 11 OUTPUT

```
pragnya@DESKTOP-AUGV0MN:~$ nano producer.c
pragnya@DESKTOP-AUGV0MN:~$ gcc producer.c -o producer -pthread
pragnya@DESKTOP-AUGV0MN:~$ ./producer
Produced: 1
Consumed: 1
Produced: 2
Consumed: 2
Produced: 3
Consumed: 3
Produced: 4
Consumed: 4
Produced: 5
Consumed: 5
Produced: 6
Consumed: 6
Produced: 7
Consumed: 7
Produced: 8
Consumed: 8
Produced: 9
Consumed: 9
Produced: 10
Consumed: 10
```